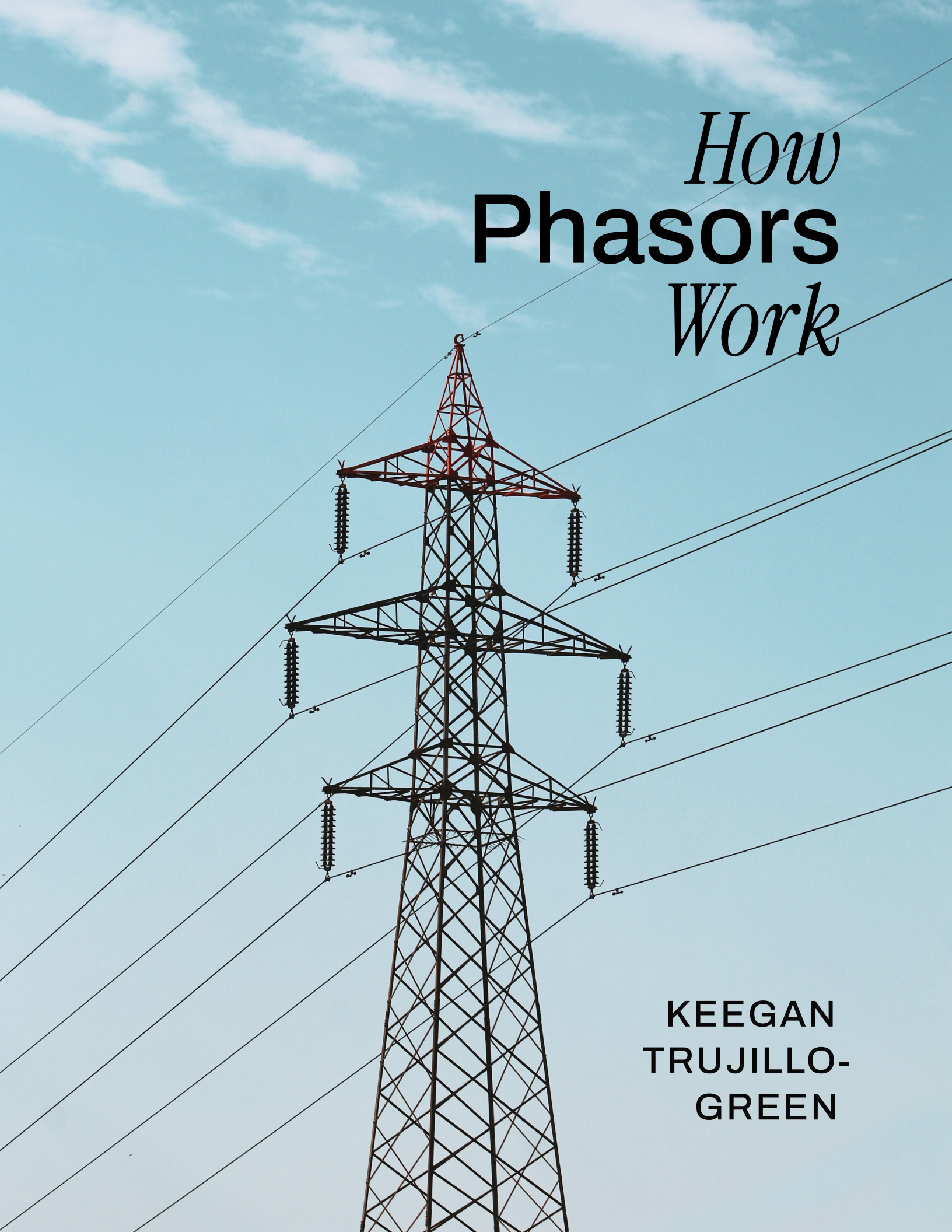




How **Phasors** *Work*

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i Note

The latest HTML version of this book is available at:

keeganmjgreen.github.io/how_phasors_work

The latest PDF version is available at:

github.com/keeganmjgreen/how_phasors_work/blob/main/how_phasors_work.pdf

INTRODUCTION

1.1 What is AC power?

Alternating current, or AC, simply describes electric current that is oscillating. However, the term is also often used to describe oscillating voltage and electric power. This is perhaps more useful, as AC voltage is what drives an AC current, and the result—AC power—is what we use to do useful work. AC means that the voltage and current oscillates a fixed number of times per second, that is, at a constant frequency f . In the vast majority of cases, this oscillation follows a sinusoidal wave.

Why do we use AC power? The key is in the name—alternating current is used in high-power applications, whether on the supply side or on the demand side.

On the supply side, we have the generation of electric power. Electric generators usually operate using rotating components. As a result, the most efficient types of generators—namely, *alternators*—produce AC power, not DC power. Historically, when DC was favored, *dynamos* were used to generate DC power, but they are less efficient. Even now when DC power is needed, alternators are used, and their output is efficiently converted to DC. Examples of alternators include the alternator in your car (or your EV's motor when used for regenerative braking), diesel generators such as those used by homeowners to power homes during power outages, and the generator in a power plant. These all produce AC power.

On the demand side, we have the consumption of electric power. The largest standalone electric loads are electric motors, the most efficient types of which consume AC power. For example, the electric motors in a diesel–electric ship or train, or in a large industrial plant, all consume AC power.

It is not necessarily enough that AC power is practical to generate and practical to use. In the case of the electrical grid, it must also be practical to transmit over long distances from where power is traditionally generated—large, centralized power plants—to the cities where it is consumed. However, transmitting power over long distances incurs power losses due to the resistance of transmission lines. Minimizing the power losses is achieved by transmitting power at high voltages. High voltages, however, are obviously unsafe and impractical for most consumers, so the voltage must be lowered as we go from transmission lines to distribution lines, into cities and neighborhoods. Furthermore, the voltage must typically be stepped up at its source—the power plant—prior to transmission. Transformers are the easiest and most efficient way to convert between voltage levels, but require AC instead of DC. So, in order to transmit power over long distances to where it is needed, while minimizing losses and delivering power at a safe and convenient voltage, AC power must be used.

1.2 Loads of all kinds

An electric circuit is only useful if it accomplishes a task. In general, that task may be concerned with logic (as in a computer), signals (as in audio equipment or radio-frequency communications), or the transmission and conversion of energy (as in the electrical grid). Discussion will focus on the electrical grid, but also applies to “off-the-grid” applications such as diesel–electric vehicles. As the previous explanation of why we use AC power indicated, being able to transmit energy from one location to another is among the strengths of AC power. However, electrical energy is useless in itself unless it can be generated in the first place and can then go on to do something useful. A different form of energy must be converted to electrical energy (whether directly or indirectly) from an energy source. For example, nuclear, hydro, gas, and solar are sources of atomic, kinetic, chemical, and light energy, respectively. In most cases, the final conversion to electrical energy takes place in an electric generator. But the ability to generate electricity is useless unless electric loads are plugged-in and consuming power; without loads, the electrical grid would technically no longer be a circuit and would serve no purpose. Thus, the electrical grid, as a circuit, includes the loads plugged into it. The electrical energy that is generated is only useful if it can be converted, once again, into another form of energy. For example, a ceiling fan converts electrical energy to mechanical energy. Even a laptop computer—which when on battery power is just a huge, standalone logic circuit—generates heat as a necessary byproduct of its operation. When plugged in, from the grid’s perspective, a computer is little more than an electric heater.

Any load that can be plugged into the grid can be modeled by one of three simple discrete components—resistor, inductor, capacitor—or a combination thereof. For example, a laptop computer can be modeled by a resistor. A ceiling fan can be modeled by a combination of resistors and inductors. Furthermore, on the supply side, an alternator can be modeled by an AC voltage source, resistor, and inductor. Being able to model any load or generator in terms of just a few simple, standard electric components means that the methods used to analyze one AC circuit can be applied just as well to another.

A resistor dissipates electrical power. It typically does so by converting electrical energy to thermal energy, but in the case of loads such as a motor, most of it is converted to something other than heat (in this case, mechanical energy). Inductors and capacitors are also able to convert electrical energy. But, unlike resistors, that conversion is bidirectional; they can store their energy and convert it back to electrical energy. An inductor stores magnetic potential energy in a magnetic field, which can be used to temporarily sustain current, even when external voltage is removed. A capacitor stores electric potential energy (and thus electric potential, a voltage) in an electric field, which can be used to temporarily sustain voltage.

1.3 DC versus AC, resistive versus reactive

Consider a circuit with a voltage source and a purely resistive load R . If the voltage source is a DC source, of constant voltage V , the circuit is DC. Current flows according to Ohm’s law, $I = V/R$. As we will see in *What is a Phasor?*, if the voltage source is AC, its voltage $v(t)$ can be expressed as $V\cos(2\pi ft + \theta)$. Ohm’s law still applies, and would now be written as $i(t) = v(t)/R$. This means that as the voltage sinusoid $v(t)$ reaches a peak, so does the current sinusoid $i(t)$ —they are *in phase*. Figure 1.1 illustrates this. However, if we were to replace the resistor with an inductor or capacitive, the current would no longer be in phase with the voltage. A capacitor’s current lags behind the AC voltage, and an inductor’s leads it. As will be discussed in *Complex Power and the Power Factor*, this is closely related to the rate at which energy is converted in inductors and capacitors.

Figure 1.1(a): Purely resistive load

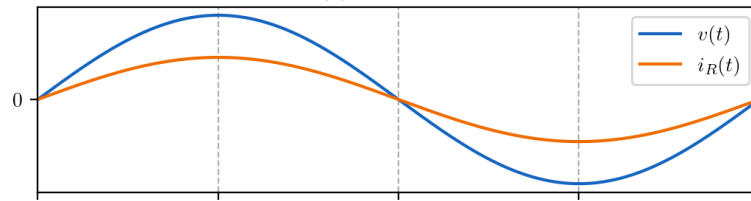


Figure 1.1(b): Purely inductive load

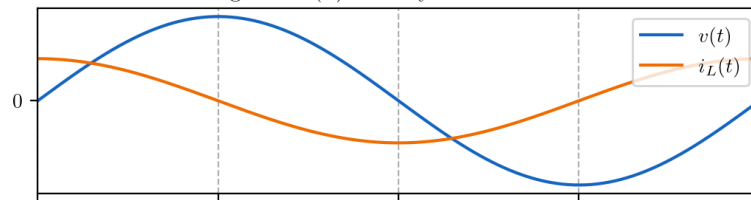
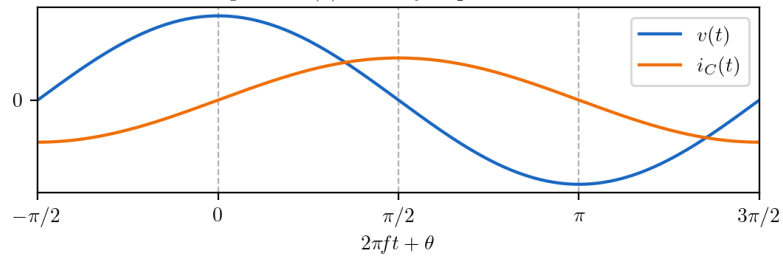


Figure 1.1(c): Purely capacitive load



WHAT IS A PHASOR?

2.1 Phasors: A notation and way to make calculations straightforward

A phasor is analogous to a negative number, in the sense that both negative numbers and phasors are mathematical representations, notations, and ways to make calculations straightforward.

A negative number is a *mathematical representation* of the opposite of a quantity. It is distinct from the number zero which represents the absence or lack of a quantity. You can have no fewer than zero apples. Similarly, you cannot have a negative quantity of dollar bills. However, if the quantity of dollars refers not to physical legal tender but to an accounting of how many units of currency you have at your disposal, then a negative quantity represents the opposite of having dollars at your disposal—that is, owing dollars. In addition to this representation, negative numbers offer:

- *A notation.* By prefixing a regular number (x) with a minus sign ($-$), it becomes negative ($-x$).
- *A way to make calculations straightforward.* A negative number is like a different kind of quantity on which arithmetic operations can be done by treating them accordingly; to subtract a negative quantity you add its positive counterpart, to multiply you multiply by the positive counterpart and negate the product, and so on.

Similarly, a phasor is a *mathematical representation* of a sinusoid, in terms of its amplitude A and phase θ . A sinusoid already has a mathematical representation in the time domain with the notation $A \cos(\omega t + \theta)$. However, while this representation makes calculations possible, it does not make them straightforward and does not have a concise notation. Representing sinusoids in the phasor domain offers:

- *A notation.* Phasors can be written concisely, such as in polar form $A \angle \theta$. The “ ωt ” is implicit.
- *A way to make calculations straightforward.* As we will see in *Life With and Without Phasors*, working in the phasor domain allows you to write and solve AC circuit analysis equations more quickly and easily than working in the time domain. Working in the time domain requires writing and solving differential equations from scratch for each circuit, even though the differential equations and their solutions are similar from one circuit to another. It turns out that the calculations that can be done with phasors are the bare minimum calculations that must be done in order to solve an AC circuit.

2.2 Back to basics: Treating sinusoids like vectors

Consider the right triangle XYZ in Figure 2.1(a). This triangle (like every triangle) has a ratio between length of side XZ (adjacent to ϕ_X) and the length of side XY (the hypotenuse). The cosine function represents the relationship between this ratio and angle ϕ_X :

$$\cos \phi_X = \frac{XZ}{XY}$$

As ϕ_X ranges from 0 to 2π radians (or 0 to 360 degrees), point Y draws a circle centered around X whose radius A is the hypotenuse's length, as in Figure 2.1(b).

If we were to plot the length of side XZ as a function of ϕ_X , we would have a cosine wave of amplitude $A = XY$, as in Figure 2.1(c), that repeats every multiple of 2π radians. This cosine wave may represent AC voltage or current.

$$A \cos \phi_X$$

If we wanted the drawing of our circle to depend on time t , and we wanted f circles to be drawn per second, we would replace the independent variable ϕ_X with t :

$$A \cos 2\pi f t$$

where f is frequency in units of cycles per second (or *Hertz*, Hz). For conciseness, we can replace $2\pi f$ with ω , the *natural frequency*, which is in units of radians per second:

$$A \cos \omega t$$

If we wanted to shift the drawing of our circle in time, we could add or subtract a constant from t . However, the mathematically equivalent convention is to add or subtract a constant θ , called the *phase*, from ωt . Now we're drawing our circle with a head start of θ , from θ to $\theta + 2\pi$ radians, as in Figure 2.1(d). This corresponds to a phase-shifted cosine wave:

$$A \cos(\omega t + \theta)$$

What if we wanted to add cosine waves? If the cosine wave represents AC voltage or current, being able to do so is important for AC circuit analysis. We can prove that the sum of two phase-shifted cosine waves is another phase-shifted cosine wave, and apply trigonometric identities to calculate it. However, the geometric approach is more intuitive. Consider that point X of the first wave's triangle is at the origin of its two-dimensional space, and that the second cosine wave is similarly formed at any moment in time by a second triangle $X'Y'Z'$. If the two waves were added, then the second triangle would be translated such that its point X' would be situated at the first triangle's point Y . Thus, the second triangle's point Y' would have the coordinates $(XZ + X'Z', YZ + Y'Z')$, representing the sum of the two waves. Similarly, if we were instead subtracting the second wave from the first, the point Y' would have the coordinates $(XZ - X'Z', YZ - Y'Z')$. Clearly, the addition and subtraction of the triangles by which cosine waves are formed satisfies the properties of vector addition and subtraction.

Figure 2.1(a)

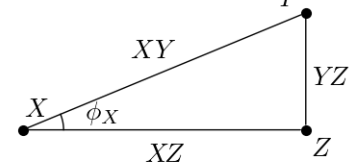


Figure 2.1(b)

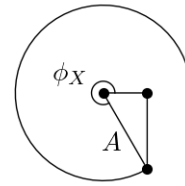


Figure 2.1(c)

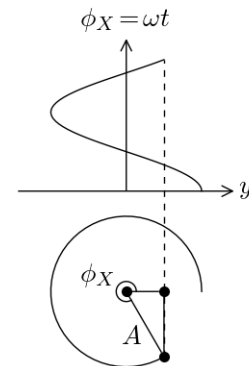


Figure 2.1(d)

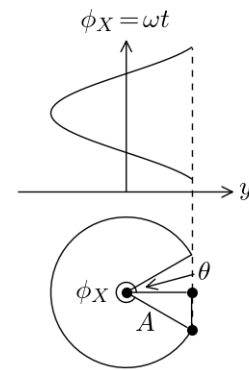
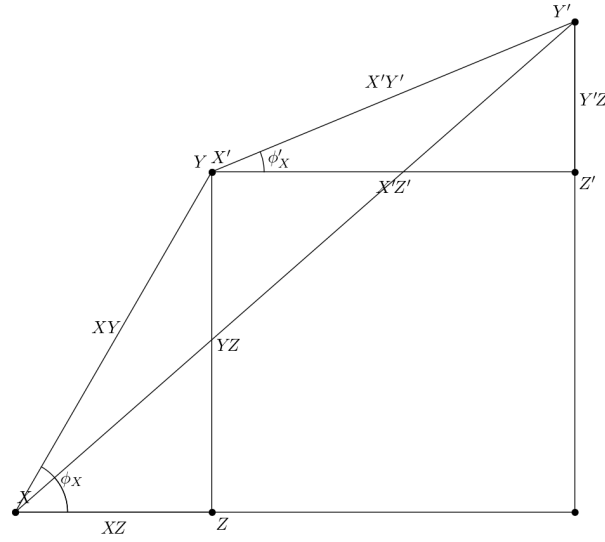


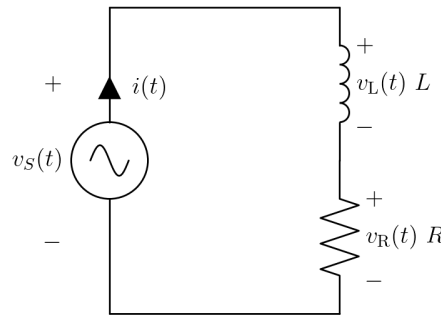
Figure 2.2



Phasors satisfy properties of vectors—addition, subtraction, multiplication by a scalar, and so on. However, phasors are not vectors. Vectors exist in n -dimensional euclidean space. As the geometric analogy suggested, phasors exist in two-dimensional space. As we will learn in the next section, the two-dimensional space in which phasors exist is not even euclidean, and phasors must satisfy additional properties that vectors do not have.

2.3 Reverse engineering the concept of phasors

Let's reverse-engineer the concept of phasors. Consider the following series RL circuit consisting of a voltage source S , a resistor R , and an inductor L :



The voltage source generates a cosine wave $v_S(t) = |V_S| \cos(\omega t + \theta_S)$. If using the geometric analogy to represent the voltage source's cosine wave, it would have coordinates $(V_{Sx}, V_{Sy}) = (|V_S| \cos \theta_S, |V_S| \sin \theta_S)$. Say that we want to solve for the steady-state current $i(t)$ through the circuit. We would start by writing the integro-differential equation:

$$\begin{aligned}
 i(t) &= \frac{1}{L} \int_0^t v_L(\tau) d\tau \\
 &= \frac{1}{L} \int_0^t (v_S(\tau) - v_R(\tau)) d\tau \\
 &= \frac{1}{L} \int_0^t (v_S(\tau) - i(\tau)R) d\tau
 \end{aligned} \tag{2.1}$$

The steady-state solution to this integro-differential equation is as follows. Note that setting up and solving AC circuits' integro-differential equations will be discussed further in [Chapter 3](#).

$$i(t) = \frac{|V_S|}{\sqrt{R^2 + (\omega L)^2}} \cos\left(\omega t + \theta_S - \arctan \frac{\omega L}{R}\right)$$

Using the geometric analogy, we can represent the sinusoid of $i(t)$ using coordinates (I_x, I_y) . We can write I_x as:

$$\begin{aligned} I_x &= \frac{|V_S|}{\sqrt{R^2 + (\omega L)^2}} \cos\left(\theta_S - \arctan \frac{\omega L}{R}\right) \\ &= \frac{|V_S|}{\sqrt{R^2 + (\omega L)^2}} \left(\cos \theta_S \cdot \cos\left(\arctan \frac{\omega L}{R}\right) + \sin \theta_S \cdot \sin\left(\arctan \frac{\omega L}{R}\right) \right) \\ &= \frac{|V_S|}{\sqrt{R^2 + (\omega L)^2}} \left(\cos \theta_S \cdot \frac{R}{\sqrt{R^2 + (\omega L)^2}} + \sin \theta_S \cdot \frac{\omega L}{\sqrt{R^2 + (\omega L)^2}} \right) \\ &= \frac{|V_S| \cos \theta_S \cdot R + |V_S| \sin \theta_S \cdot \omega L}{R^2 + (\omega L)^2} \end{aligned} \quad (2.2)$$

Similarly, I_y can be written as follows.

$$\begin{aligned} I_y &= \frac{|V_S|}{\sqrt{R^2 + (\omega L)^2}} \sin\left(\theta_S - \arctan \frac{\omega L}{R}\right) \\ &= \frac{|V_S| \sin \theta_S \cdot R - |V_S| \cos \theta_S \cdot \omega L}{R^2 + (\omega L)^2} \end{aligned} \quad (2.3)$$

Let's substitute V_{Sx} for $|V_S| \cos \theta_S$ and V_{Sy} for $|V_S| \sin \theta_S$. Let's also replace R and ωL with new quantities Z_x and Z_y , whose significance will become apparent shortly.

$$(I_x, I_y) = \left(\frac{V_{Sx}Z_x + V_{Sy}Z_y}{Z_x^2 + Z_y^2}, \frac{V_{Sy}Z_x - V_{Sx}Z_y}{Z_x^2 + Z_y^2} \right)$$

Making use of the properties of complex numbers (in the box below), we discover what is important about being able to write (I_x, I_y) in this way. The complex number $I_x + jI_y$ (where j is the imaginary unit) is the result of the following complex number division:

$$I_x + jI_y = \frac{V_{Sx} + jV_{Sy}}{Z_x + jZ_y} \quad (2.4)$$

Properties of complex numbers

Addition and subtraction:

$$(a + jb) \pm (c + jd) = (a \pm c) + j(b \pm d)$$

Multiplication:

$$(a + jb)(c + jd) = ac + jda + jbc + (jb)(jd) = (ac - bd) + j(bc + ad)$$

Division:

$$\begin{aligned} \frac{a + jb}{c + jd} &= \frac{a + jb}{c + jd} \cdot \frac{c - jd}{c - jd} \quad (\text{Multiply both numerator and denominator by denominator's conjugate.}) \\ &= \frac{ac - jda + jbc - (jb)(jd)}{c^2 - jdc + jdc - (jd)(jd)} \\ &= \frac{(ac + bd) + j(bc - ad)}{c^2 + d^2} \end{aligned}$$

Knowing that $Z_x = R$, Equation (2.4) feels like Ohm's law. If we were to remove the inductor L from the circuit and thus set $Z_y = \omega L$ to zero, the equation indeed becomes Ohm's law for a purely resistive circuit:

$$\begin{aligned} I_x + j I_y &= \frac{V_{Sx} + j V_{Sy}}{R} = \frac{|V_S| \cos \theta_S}{R} + j \frac{|V_S| \sin \theta_S}{R} \\ \Rightarrow (I_x, I_y) &= \left(\frac{|V_S| \cos \theta_S}{R}, \frac{|V_S| \sin \theta_S}{R} \right) \\ \Rightarrow i(t) &= \frac{v_S(t)}{R} \end{aligned}$$

In fact, Equation (2.4) is the generalization of Ohm's law for an AC circuit with inductive and capacitive components, and $Z_x + j Z_y$, called *impedance*, is the generalization of electrical resistance R for such a circuit. Just as we could write $i(t) = v_S(t)/R$ for a purely resistive AC circuit, we can write $I_x + j I_y = (V_{Sx} + j V_{Sy})/(Z_x + j Z_y)$ for an AC circuit with inductive and capacitive components.

This indicates to us that the coordinates we've been working with in our geometric analogy are actually complex numbers. They satisfy all the properties of complex numbers, including addition, subtraction, multiplication by a scalar, multiplication, and division (as we've just seen). A complex number, as used to represent a sinusoidal voltage or current, is what we call a phasor. Furthermore, a complex number, as used to represent the ratio between a voltage phasor and a current phasor, is what we call impedance. Impedance, however, is not itself a phasor because it does not represent a sine wave.

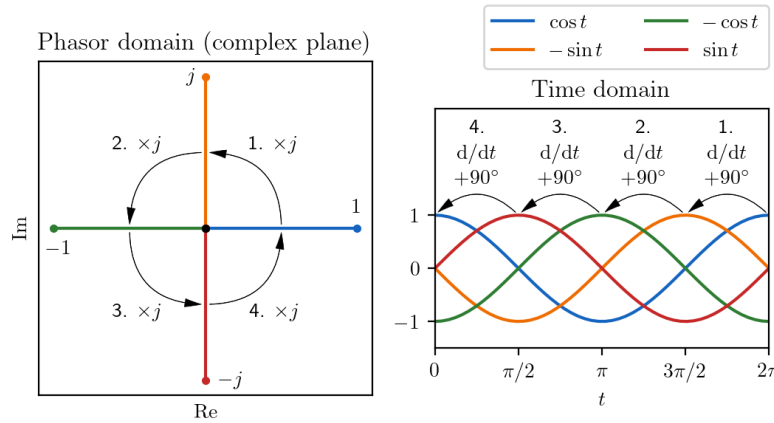
The most important feature of phasors is that they allow us to do calculations using complex numbers rather than solving differential equations. For example, rather than solving differential equation (2.1), we can quickly convert $v_S(t)$ to a phasor, solve a regular equation in the phasor domain, and convert the solution to $i(t)$ back in the time domain.

You may be thinking that complex numbers are an unintuitive or downright far-fetched way to represent solving differential equations. In answer to this, consider that the behavior of multiplying a complex number of unit magnitude by j is identical to the behavior of differentiating a sinusoid of unit amplitude with respect to time, as shown in the following diagram. Similarly, by reversing the diagram, the behavior of dividing a complex number of unit magnitude by j is identical to the behavior of integrating a sinusoid of unit amplitude with respect to time. So, instead of writing derivatives and integrals to model capacitors and inductors, we can simply multiply and divide by j .

	1	$\cos t$	
Multiply by j	$\downarrow$	$\downarrow$	Differentiate w.r.t. t
	j	$-\sin t$	
Multiply by j	$\downarrow$	$\downarrow$	Differentiate w.r.t. t
	-1	$-\cos t$	
Multiply by j	$\downarrow$	$\downarrow$	Differentiate w.r.t. t
	$-j$	$\sin t$	
Multiply by j	$\downarrow$	$\downarrow$	Differentiate w.r.t. t
	1	$\cos t$	

Notice too how there is a $+\pi/2$ (or $+90^\circ$) phase shift from $\cos t$ to $-\sin t$, and from $-\sin t$ to $-\cos t$, and so on. Thus, multiplying or dividing by j in the phasor domain is also equivalent to shifting the corresponding sinusoid in the time domain by a quarter cycle.

Figure 2.4



2.4 Notation for phasors

Previously, we treated phasors as coordinates in two-dimensional space, with x and y components. For example, we had:

$$(V_{Sx}, V_{Sy}) \quad (I_x, I_y)$$

However, now that we know that phasors (and impedances) are just complex numbers, we can replace this notation with complex number notation, with a real and imaginary component. For example:

$$V_S = \text{Re}(V_S) + j \text{Im}(V_S) \quad I = \text{Re}(I) + j \text{Im}(I) \quad Z = \text{Re}(Z) + j \text{Im}(Z)$$

2.4.1 Polar form

Let's introduce a polar form of complex number notation, which is used to write a phasor (or impedance) in terms of amplitude and phase, rather than in terms of its real and imaginary components. For example, using Euler's formula $e^{jx} = \cos x + j \sin x$, we can write $V_S = |V_S| \cos \theta_S + j |V_S| \sin \theta_S$ as:

$$|V_S| e^{j\theta_S}$$

Or using *Steinmetz notation*:

$$|V_S| \angle \theta_S$$

We can just as well apply this polar form to equations (2.2) and (2.3):

$$\begin{aligned} I &= \frac{|V_S|}{\sqrt{R^2 + (\omega L)^2}} \cos\left(\theta_S - \arctan \frac{\omega L}{R}\right) + j \frac{|V_S|}{\sqrt{R^2 + (\omega L)^2}} \sin\left(\theta_S - \arctan \frac{\omega L}{R}\right) \\ &= \frac{|V_S|}{\sqrt{R^2 + (\omega L)^2}} \angle \left(\theta_S - \arctan \frac{\omega L}{R}\right) \end{aligned} \quad (2.5)$$

Previously, equations (2.2) and (2.3) led us to see a way that I can be calculated from V_S by dividing by impedance Z using division of complex numbers. If we take a close look at Equation (2.5), we can find an additional, equivalent way in which I can be calculated from V_S . Notice how V_S can be converted to I by dividing its amplitude $|V_S|$ by $\sqrt{R^2 + (\omega L)^2}$ and by subtracting $\arctan(\omega L/R)$ from θ_S . This demonstrates a useful property of phasors and complex numbers in general in polar form—two complex numbers can easily be divided by dividing their amplitudes and subtracting their phases:

$$\frac{A_1 \angle \theta_1}{A_2 \angle \theta_2} = \frac{A_1}{A_2} \angle (\theta_1 - \theta_2)$$

Of course, all properties of complex numbers can be written in polar form, but multiplying and dividing complex numbers is easiest when they are in polar form. On the other hand, adding and subtracting complex numbers is easiest when they are in the non-polar form.

i Properties of complex numbers (polar form)

Addition and subtraction:

$$A_1 \angle \theta_1 \pm A_2 \angle \theta_2 = \sqrt{(A_1 \cos \theta_1 \pm A_2 \cos \theta_2)^2 + (A_1 \sin \theta_1 \pm A_2 \sin \theta_2)^2} \angle \arctan \frac{A_1 \sin \theta_1 \pm A_2 \sin \theta_2}{A_1 \cos \theta_1 \pm A_2 \cos \theta_2}$$

Multiplication:

$$A_1 \angle \theta_1 \cdot A_2 \angle \theta_2 = A_1 A_2 \angle (\theta_1 + \theta_2)$$

Division:

$$\frac{A_1 \angle \theta_1}{A_2 \angle \theta_2} = \frac{A_1}{A_2} \angle (\theta_1 - \theta_2)$$

Complex numbers can be converted to and from polar form using the following formulae:

$$a + jb = \sqrt{a^2 + b^2} \angle \arctan \frac{b}{a}$$

$$A \angle \theta = A \cos \theta + j A \sin \theta$$

2.5 The phasor transformation

Thus far, we've looked at how phasors fit into AC circuit mathematics to see where they come from. At this point we can provide a formal definition for a phasor and how it is obtained from a sinusoid in the time domain. Consider the sinusoid $A \cos(\omega t + \theta)$. Adding an imaginary component $j A \sin(\omega t + \theta)$ gives us:

$$A \cos(\omega t + \theta) + j A \sin(\omega t + \theta)$$

By setting $t = 0$ and thus removing the time-varying part ωt , we can convert this to a phasor as follows. Setting t to a specific value allows us to produce a "snapshot" of the sinusoid. Any specific value for t will do, however $t = 0$ is most convenient.

$$A \cos \theta + j \sin \theta$$

If we convert to polar form,

$$A e^{j(\omega t + \theta)} = A e^{j\omega t} e^{j\theta},$$

we can accomplish the same thing by dividing by the time-varying factor $e^{j\omega t}$ to obtain:

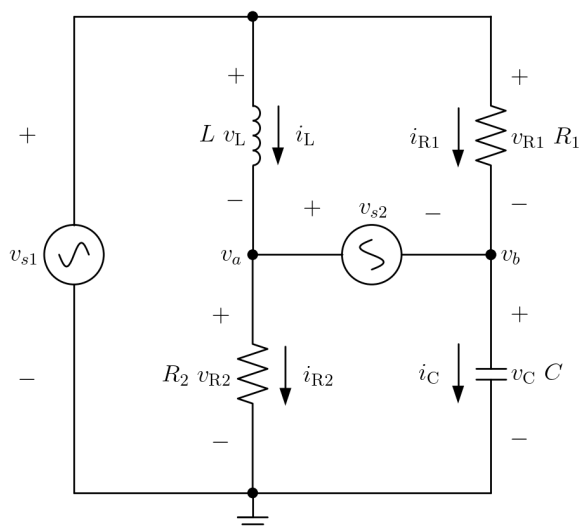
$$A e^{j\theta}$$

LIFE WITH AND WITHOUT PHASORS

One of the main reasons for many people being confused about phasors is that phasors are abstract, but they don't understand why the abstraction exists. In other words, they may know what phasors *represent*, but they don't know what they *replace*—they don't know of any alternative approach for analyzing AC circuits that can be used to put phasors in perspective. This is understandable as many electrical engineering textbooks poorly explain why we use phasors.

With the aid of an example AC circuit, this chapter goes over what's hidden behind the phasor abstraction. By first trying to solve the circuit without phasors, we will better understand how AC circuit analysis works at its core. And by then solving the circuit with phasors, we will see that it is not only a more concise notation, but also a simpler mathematical representation and a faster method for analyzing AC circuits.

The following AC circuit is our example circuit. Note that it has multiple AC sources and an bridge-type topology that does not allow us to consolidate impedances using formulae for impedances in series or in parallel.



Say that we want to solve for voltages $v_a(t)$ and $v_b(t)$. Doing so would allow us to solve for the voltage across—and therefore the current through—any one component. Just as for a purely resistive DC network, Kirchhoff's Current Law (KCL) applies and we can do [nodal analysis](#) to determine the voltages at nodes a and b . In particular we must use the [supernode technique](#) with a single node ab because the current through voltage source v_{s2} cannot be put in terms of $v_a(t)$ or $v_b(t)$. This loses us an equation, but we gain one in its place with the following relationship:

$$v_{s2}(t) = v_a(t) - v_b(t) \quad (3.1)$$

Let's try solving the circuit using node voltage analysis, with and without phasors.

Note

While the parameters $|V_{s1}|$, ω , R_1 , etc. can take on any numerical values, at certain points in the chapter we will substitute the following very arbitrary values to simplify or be able to plot our equations.

$$\begin{aligned} |V_{s1}| &= 1 \text{ V} & |V_{s2}| &= 2 \text{ V} & \omega &= 3 \text{ rad/s} \\ R_1 &= 4 \Omega & R_2 &= 5 \Omega & L &= 6 \text{ H} & C &= 7 \text{ F} \end{aligned} \quad (3.2)$$

3.1 Solving the circuit without phasors

Using KCL, the sum of currents flowing out of supernode ab must sum to zero. We can therefore write:

$$-i_L(t) + i_{R2}(t) - i_{R1}(t) + i_C(t) = 0$$

Substituting in the components' characteristic equations gives us:

$$-\frac{1}{L} \int_0^t v_L(\tau) d\tau + \frac{v_{R2}(t)}{R_2} - \frac{v_{R1}(t)}{R_1} + C \frac{dv_C(t)}{dt} = 0$$

Putting everything in terms of the node voltages yields:

$$-\frac{1}{L} \int_0^t (v_{s1}(\tau) - v_a(\tau)) d\tau + \frac{v_a(t)}{R_2} - \frac{v_{s1}(t) - v_b(t)}{R_1} + C \frac{dv_b(t)}{dt} = 0 \quad (3.3)$$

Paired with Equation (3.1), this represents a system of two integro-differential equations. By differentiating, we put this in the more familiar form of a system of differential equations (in which the highest-order derivative is two):

$$\begin{aligned} -\frac{1}{L} (v_{s1}(t) - v_a(t)) + \frac{1}{R_2} \frac{dv_a(t)}{dt} - \frac{1}{R_1} \frac{d}{dt} (v_{s1}(t) - v_b(t)) + C \frac{d^2 v_b(t)}{dt^2} &= 0 \\ v_{s2}(t) &= v_a(t) - v_b(t) \end{aligned}$$

We put this into a canonical form of a system of first-order differential equations by introducing an auxiliary variable $v'_b(t)$ and by using Equation (3.1) to replace the need for variable $v_a(t)$ with $v'_b(t)$:

$$\begin{aligned} \frac{dv_b(t)}{dt} &= v'_b(t) \\ \frac{dv'_b(t)}{dt} &= \frac{1}{C} \left[\frac{v_{s1}(t) - (v_{s2}(t) + v_b(t))}{L} - \frac{v'_{s2}(t) + v'_b(t)}{R_2} + \frac{v'_{s1}(t) - v'_b(t)}{R_1} \right] \end{aligned}$$

We specify that $v_{si}(t) = V_{si} \sin \omega t$ (and $v'_{si}(t) = \omega V_{si} \cos \omega t$):

$$\begin{aligned} \frac{dv_b(t)}{dt} &= v'_b(t) \\ \frac{dv'_b(t)}{dt} &= \frac{1}{C} \left[\frac{(|V_{s1}| - |V_{s2}|) \sin \omega t - v_b(t)}{L} - \frac{\omega |V_{s2}| \cos \omega t + v'_b(t)}{R_2} + \frac{\omega |V_{s1}| \cos \omega t - v'_b(t)}{R_1} \right] \end{aligned}$$

Our system is now in terms of two unknowns— $v_b(t)$ and $v'_b(t)$ —two functions to solve for. We start to solve our system of differential equations by assuming a solution of the form:

$$\begin{aligned} v_b(t) &= V_b \cos(\omega t + \theta_b) \\ v'_b(t) &= -\omega V_b \sin(\omega t + \theta_b) \end{aligned}$$

By substituting into our system of differential equations, we convert it to a regular system of equations—or, in this case, just one equation:

$$-\omega^2 V_b \cos(\omega t + \theta_b) = \frac{1}{C} \left[\frac{(|V_{s1}| - |V_{s2}|) \sin \omega t - V_b \cos(\omega t + \theta_b)}{L} + \omega \left[\left(\frac{|V_{s1}|}{R_1} - \frac{|V_{s2}|}{R_2} \right) \cos \omega t + \left(\frac{1}{R_1} + \frac{1}{R_2} \right) V_b \sin(\omega t + \theta_b) \right] \right]$$

This equation is in terms of two unknowns— V_b and θ_b . While having only one equation may appear to be insufficient to solve for two unknowns, the equation is parameterized by t and must be satisfied for all values of t . So, in this sense, we actually have an infinite number of equations at our disposal. We can pick two ‘easy’ values of t that result in two relatively simple equations:

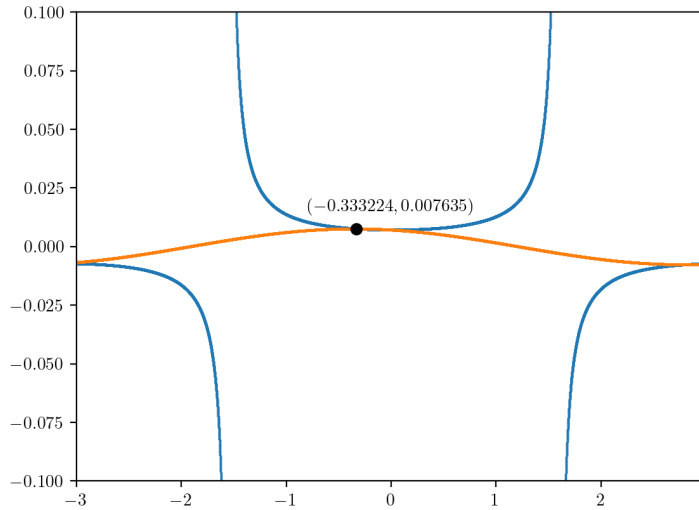
$t = 0$:

$$-\omega^2 C V_b \cos \theta_b = \frac{-V_b \cos \theta_b}{L} + \omega \left[\left(\frac{|V_{s1}|}{R_1} - \frac{|V_{s2}|}{R_2} \right) + \left(\frac{1}{R_1} + \frac{1}{R_2} \right) V_b \sin \theta_b \right]$$

$t = -\theta_b/\omega$:

$$-\omega^2 C V_b = \frac{-(|V_{s1}| - |V_{s2}|) \sin \theta_b - V_b}{L} + \omega \left(\frac{|V_{s1}|}{R_1} - \frac{|V_{s2}|}{R_2} \right) \cos \theta_b$$

This nonlinear system is inconvenient to solve. We can plot these two equations as follows and solve the system numerically. However, numerical methods are less efficient than being able to directly calculate the solution. We should be able to solve AC circuits very quickly.



Note

There are technically an infinite number of solutions for θ_b , because any multiple of 2π can be added to (or subtracted from) θ_b , resulting in an identical sinusoid with a different equation.

Furthermore, as we have seen, the process to derive the system of equations is long and notation-heavy despite being consistent between different AC circuits.

Before we see how the circuit can be solved with phasors, let's see how we can solve its system of integro-differential equations with the aid of the Laplace transform.

3.2 Solving the circuit with the Laplace transform

The Laplace transform $\mathcal{L}$ converts a function $f(t)$ (in the *time domain*) into a function $F(s)$ (in the *complex frequency domain*) through the following integration:

$$F(s) = \mathcal{L}\{f\}(s) = \int_0^{\infty} f(t) e^{-st} dt \quad (3.4)$$

The Laplace transform is closely related to the Fourier transform. Both transforms convert to the frequency domain. However, the Laplace transform converts to a frequency domain whose variable s is a *complex number*, which can simplify calculations. This domain is known as the *s-domain* or *Laplace domain*.

The Laplace transform has a number of favorable properties. One such property is that differentiating with respect to t in the time domain becomes multiplication by s in the Laplace domain. Similarly, integrating with respect to t in the time domain becomes division by s in the Laplace domain. Because of this, the Laplace transform is often used to convert a system of differential equations into a regular system of equations which can be solved more easily.

Furthermore, many useful functions in the time domain become simple, rational functions in the Laplace domain, for example:

Sine wave:	$\mathcal{L}\{\sin(\omega t) u(t)\} = \frac{\omega}{s^2 + \omega^2}$
Phase-shifted cosine wave:	$\mathcal{L}\{\cos(\omega t + \theta) u(t)\} = \frac{s \cos \theta - \omega \sin \theta}{s^2 + \omega^2}$
Exponential decay:	$\mathcal{L}\{e^{-\alpha t} u(t)\} = \frac{1}{s + \alpha}$
Exponentially decaying cosine wave:	$\mathcal{L}\{e^{-\alpha t} \cos(\omega t) u(t)\} = \frac{s + \alpha}{(s + \alpha)^2 + \omega^2}$

Listings of the Laplace transforms of common functions, such as those above, are called Laplace tables.

$u(t)$ is the Heaviside step function, which is 1 for $t \geq 0$ and 0 otherwise. The time-domain functions are multiplied by $u(t)$ —that is, only nonzero for $t \geq 0$, for the same reason that the lower bound of integration in Equation (3.4) is 0. The Laplace transform only looks at $t \geq 0$, and is thus useful for solving differential equations involving functions that are only nonzero for $t \geq 0$. This aspect allows us to answer questions about a system—particularly an electric circuit—that is modeled by differential equations, such as: “what happens when the system is turned on?”, “what happens when the value of this component suddenly changes?”, and “what happens when the circuit’s topology changes in this way?” To consider functions that are nonzero for $t < 0$, and thus model no sudden change at $t = 0$, the *two-sided* or *bilateral* Laplace transform can be used.

We want to solve the system of two integro-differential equations (3.1) and (3.3) that we derived in the previous section. That system was in terms of $v_a(t)$ and $v_b(t)$, in the time domain. First, we will apply the Laplace transform to convert it to a regular system of equations in terms of $V_a(s)$ and $V_b(s)$, in the Laplace domain. Then, we will be able to solve for $V_a(s)$ and $V_b(s)$ very easily. Finally, we will convert our solution back to the time domain using the inverse Laplace transform.

Applying the Laplace transform to equations (3.1) and (3.3) gives us:

$$-\frac{|V_{s1}|(s) - V_a(s)}{Ls} + \frac{V_a(s)}{R_2} - \frac{|V_{s1}|(s) - V_b(s)}{R_1} + CsV_b(s) = 0$$

$$|V_{s2}|(s) = V_a(s) - V_b(s)$$

Here, $|V_{s1}|(s)$ and $|V_{s2}|(s)$ are the Laplace transforms of $v_{s1}(t)$ and $v_{s2}(t)$ —not to be confused with the amplitudes $|V_{s1}|$ and $|V_{s2}|$ of $v_{s1}(t)$ and $v_{s2}(t)$.

Solving for $V_a(s)$ in the second equation and substituting it into the first equation yields:

$$-\frac{|V_{s1}|(s) - (|V_{s2}|(s) + V_b(s))}{Ls} + \frac{|V_{s2}|(s) + V_b(s)}{R_2} - \frac{|V_{s1}|(s) - V_b(s)}{R_1} + CsV_b(s) = 0$$

Solving for $V_b(s)$ gives us:

$$V_b = \frac{R_2(R_1 + Ls)|V_{s1}|(s) - R_1(R_2 + Ls)|V_{s2}|(s)}{R_1R_2 + (R_1 + R_2)Ls + R_1R_2LCs^2}$$

Substituting in $V_{si}(s) = \mathcal{L}\{V_{si} \sin(\omega t) u(t)\} = V_{si} \omega / (s^2 + \omega^2)$ yields the following. Note that, because the voltage sources are multiplied by $u(t)$, we are essentially answering the question of how the circuit responds when turned on at $t = 0$.

$$V_b = \frac{R_2(R_1 + Ls)|V_{s1}|\omega - R_1(R_2 + Ls)|V_{s2}|\omega}{(s^2 + \omega^2)(R_1R_2 + (R_1 + R_2)Ls + R_1R_2LCs^2)}$$

At this point, the algebra will be significantly easier if we substitute in known values for our parameters. Let's introduce numerical values (3.2), after which we will omit units for brevity. However, $V_b(s)$ is still in units of volts and s is still, like ω , in units of radians per second.

$$\begin{aligned} V_b(s) &= \frac{(5 \Omega)(4 \Omega + (6 \text{ H})s)(1 \text{ V})(3 \text{ rad/s}) - (4 \Omega)(5 \Omega - (6 \text{ H})s)(2 \text{ V})(3 \text{ rad/s})}{(3 \text{ rad/s})(-(4 \Omega)(5 \Omega) + (4 \Omega - 5 \Omega)(6 \text{ H})s + (4 \Omega)(5 \Omega)(6 \text{ H})(7 \text{ F})s^2)} \\ &= \frac{-54s - 60}{(s^2 + 9)(840s^2 + 54s + 20)} \end{aligned}$$

We need to convert this solution for $V_b(s)$ back to the time domain. Converting a function from the Laplace domain to the time domain can be done by looking it up in a Laplace table. However, $V_b(s)$ is too complicated in its current form to find a matching function in a Laplace table—in particular, because the order of the polynomial in the denominator is four. This is higher than the highest-order denominator of two that can be found in typical Laplace tables. We can handle this by rewriting our solution as a linear combination of simpler functions that we will be able to find in the Laplace table. To determine such simpler functions, we factor the denominator then apply *partial fraction decomposition*. The denominator needs to be split into factors whose order is at most two. This is already the case. Next, we assume the following modified format of our V_b solution and find the values A_1 through A_4 that make it true:

$$\begin{aligned} V_b(s) &= \frac{A_1s + A_2}{s^2 + 9} + \frac{A_3s + A_4}{840s^2 + 54s + 20} \\ \Rightarrow &\frac{-54s - 60}{(s^2 + 9)(840s^2 + 54s + 20)} = \frac{A_1s + A_2}{s^2 + 9} + \frac{A_3s + A_4}{840s^2 + 54s + 20} \\ \Rightarrow &-54s - 60 = (A_1s + A_2)(840s^2 + 54s + 20) + (A_3s + A_4)(s^2 + 9) \\ \Rightarrow &(-840A_1 - A_3)s^3 \\ &+ (-54A_1 - 840A_2 - A_4)s^2 \\ &+ (-20A_1 - 54A_2 - 9A_3 - 54)s \\ &+ (-20A_2 - 9A_4 - 60) \\ &= 0 \\ \Rightarrow &\begin{cases} -840A_1 - A_3 = 0 \\ -54A_1 - 840A_2 - A_4 = 0 \\ -20A_1 - 54A_2 - 9A_3 - 54 = 0 \\ -20A_2 - 9A_4 - 60 = 0 \end{cases} \\ \Rightarrow &\begin{cases} A_1 = 102600/14219461 \\ A_2 = 106539/14219461 \\ A_3 = -86184000/14219461 \\ A_4 = -95033160/14219461 \end{cases} \end{aligned}$$

Substituting A_1 through A_4 gives us:

$$V_b(s) = \underbrace{\frac{\frac{102600}{14219461}s + \frac{106539}{14219461}}{s^2 + 9}}_{\text{First term}} + \underbrace{\frac{-\frac{86184000}{14219461}s - \frac{95033160}{14219461}}{840s^2 + 54s + 20}}_{\text{Second term}}$$

Inverse Laplace transform of the first term. The first term matches the following Laplace table function:

$$\mathcal{L}\{\cos(\omega t + \theta) u(t)\} = \frac{s \cos \theta + \omega \sin \theta}{s + \omega^2}$$

Therefore:

$$\text{First term} = \frac{\frac{102600}{14219461}s + \frac{106539}{14219461}}{s^2 + 9} = B_1 \frac{s \cos \theta_1 - \omega_1 \sin \theta_1}{s + \omega_1^2}$$

The ω_1 here is (not coincidentally) the same as our natural frequency $\omega = 3$ rad/s. Now, we need to solve for θ_1 and B_1 , which we can do by setting up the following system of equations:

$$\begin{aligned} B_1 \cos \theta_1 &= 102600/14219461 \\ -3B_1 \sin \theta_1 &= 106539/14219461 \end{aligned}$$

We can solve for θ_1 by dividing the second equation by the first:

$$\frac{106539/14219461}{102600/14219461} = -3 \tan \theta_1 \Rightarrow \theta_1 = \arctan\left(-\frac{35513}{102600}\right) = -19.092^\circ$$

And we can solve for B_1 by dividing the second equation by -3 and adding its square to the square of the first:

$$\begin{aligned} \left(\frac{102600}{14219461}\right)^2 + \left(\frac{106539}{-3 \cdot 14219461}\right)^2 &= B_1^2 \cos^2 \theta_1 + B_1^2 \sin^2 \theta_1 = B_1^2 \\ \Rightarrow B_1 &= \sqrt{\left(\frac{102600}{14219461}\right)^2 + \left(\frac{35513}{14219461}\right)^2} = 0.007635 \end{aligned}$$

Therefore, the inverse Laplace transform of the first term is:

$$\mathcal{L}^{-1}\left\{\frac{\frac{102600}{14219461}s + \frac{106539}{14219461}}{s^2 + 9}\right\} = 0.007635 \cos(3t - 19.092^\circ) u(t)$$

Derivation: Laplace transform of an exponentially decaying, phase-shifted cosine wave

$$\begin{aligned} \mathcal{L}\{e^{-\alpha t} \cos(\omega t + \theta) u(t)\} &= \mathcal{L}\{e^{-\alpha t} [\cos \omega t \cos \theta - \sin \omega t \sin \theta] u(t)\} \\ &= \cos \theta \mathcal{L}\{e^{-\alpha t} \cos \omega t u(t)\} - \sin \theta \mathcal{L}\{e^{-\alpha t} \sin \omega t u(t)\} \\ &= \cos \theta \frac{s + \alpha}{(s + \alpha)^2 + \omega^2} - \sin \theta \frac{\omega}{(s + \alpha)^2 + \omega^2} \\ &= \frac{(s + \alpha) \cos \theta - \omega \sin \theta}{(s + \alpha)^2 + \omega^2} \end{aligned} \quad (3.5)$$

Inverse Laplace transform of the second term. The second term matches the following function from (3.5):

$$\mathcal{L}\{e^{-\alpha t} \cos(\omega t + \theta) u(t)\} = \frac{(s + \alpha) \cos \theta - \omega \sin \theta}{(s + \alpha)^2 + \omega^2}$$

Therefore:

$$\begin{aligned} \text{Second term} &= \frac{-\frac{86184000}{14219461}s - B_2 \frac{95033160}{14219461}}{840s^2 + 54s + 20} = \frac{(s + \alpha_2) \cos \theta_2 - \omega_2 \sin \theta_2}{(s + \alpha_2)^2 + \omega_2^2} \\ \Rightarrow \frac{-\frac{102600}{14219461}s - \frac{791943}{99536227}}{s^2 + \frac{9}{140}s + \frac{1}{42}} &= B_2 \frac{s \cos \theta_2 + (\alpha_2 \cos \theta_2 - \omega_2 \sin \theta_2)}{s^2 + 2\alpha_2 s + (\alpha_2^2 + \omega_2^2)} \end{aligned}$$

We need to solve for α_2 , ω_2 , θ_2 , and B_2 :

$$\begin{aligned} 9/140 &= 2\alpha_2 \Rightarrow \alpha_2 = 9/280 = 0.03214 \\ \alpha_2^2 + \omega_2^2 &= \frac{1}{42} \Rightarrow \omega_2 = \sqrt{\frac{1}{42} - \left(\frac{9}{280}\right)^2} = 0.15092 \text{ rad/s} \end{aligned}$$

We can solve for θ_2 and B_2 by setting up and solving the following system of equations:

$$\begin{aligned} \left. \begin{aligned} -102600/14219461 &= B_2 \cos \theta_2 \\ -791943/99536227 &= B_2 (0.03214 \cos \theta_2 - 0.15092 \sin \theta_2) \end{aligned} \right\} \Rightarrow \begin{cases} \theta_2 = -81.9756^\circ \\ B_2 = -0.05169 \end{cases} \\ B_2 &= -\frac{102600}{14219461 \cos \theta_2} \end{aligned}$$

Therefore, the inverse Laplace transform of the second term is:

$$\mathcal{L}^{-1} \left\{ \frac{-\frac{86184000}{14219461}s - \frac{95033160}{14219461}}{840s^2 + 54s + 20} \right\} = -0.05169 e^{-0.03214t} \cos(0.15092t - 81.9756^\circ) u(t)$$

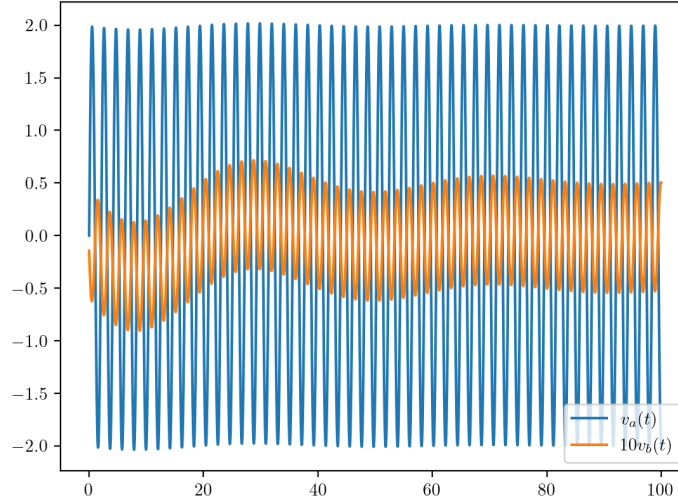
And finally, substituting in the inverse Laplace transforms of the first and second terms yields the following full solution for $v_b(t)$ in the time domain:

$$v_b(t) = \underbrace{(0.007635 \cos(3t - 19.092^\circ))}_{\text{Steady-state response, } v_b^{SS}(t)} - \underbrace{0.05169 e^{-0.03214t} \cos(0.15092t - 81.9756^\circ)}_{\text{Transient response}} u(t)$$

From this we can calculate $v_a(t)$ via trigonometric identities. Its transient response is the same as for $v_b(t)$, and its steady-state response is:

$$\begin{aligned} v_a^{SS}(t) &= v_{s2}(t) + v_b^{SS}(t) \\ &= 0.007635 \cos(3t - 19.092^\circ) + 2 \sin(3t) \\ &= 1.99752 \cos(3t - 89.7930^\circ) \end{aligned}$$

The solution for $v_b(t)$ (and $v_a(t)$) has two distinct terms. It is a linear combination of cosine waves, each with their own natural frequency and phase, but one of the cosine waves is exponentially decaying. The exponentially decaying cosine models how the circuit responds when turned on at $t = 0$ —its *transient response*. The non-decaying cosine is how the circuit responds in the longer term—its *steady-state response* as t tends to infinity (but practically much sooner as the transient quickly dies out).



3.3 Solving the circuit with phasors

Using KCL, the sum of currents flowing out of supernode ab must sum to zero. We can therefore write:

$$-I_L + I_{R2} - I_{R1} + I_C = 0$$

$$-\frac{V_L}{Z_L} + \frac{V_{R2}}{Z_{R2}} - \frac{V_{R1}}{Z_{R1}} + \frac{V_C}{Z_C} = 0$$

Substituting in the components' impedances and putting everything in terms of the node voltages gives us:

$$-\frac{V_{s1} - V_a}{j\omega L} + \frac{V_a}{R_2} - \frac{V_{s1} - V_b}{R_1} + \frac{V_b}{j\frac{1}{\omega C}} = 0$$

Paired with Equation (3.1), $V_a - V_b = V_{s2}$, this represents a system of two linear equations in two unknowns V_a and V_b . Substituting Equation (3.1) into the above yields:

$$j\frac{V_{s1} - (V_{s2} + V_b)}{\omega L} + \frac{V_{s2} + V_b}{R_2} - \frac{V_{s1} - V_b}{R_1} + j\omega C V_b = 0$$

Solving for V_b yields the following, and we can solve for V_a from this using Equation (3.1).

$$V_b = \frac{\omega L(R_2 V_{s1} - R_1 V_{s2}) + j R_1 R_2 (V_{s2} - V_{s1})}{\omega L(R_1 + R_2) + j(\omega^2 C L - 1) R_1 R_2}$$

Getting here required some algebra involving complex numbers. However, when solving a real circuit whose parameters have known values, this process would have been aided significantly by being able to evaluate/simplify expressions at each

step. Let's introduce numerical values (3.2) to see how V_b and V_a can be evaluated.

$$\begin{aligned}
 V_b &= \frac{(3 \text{ rad/s})(6 \text{ H})((5 \Omega)(j 1 \text{ V}) - (4 \Omega)(j 2 \text{ V})) + j(4 \Omega)(5 \Omega)((j 2 \text{ V}) - (j 1 \text{ V}))}{(3 \text{ rad/s})(6 \text{ H})((4 \Omega) + (5 \Omega)) + j((3 \text{ rad/s})^2(7 \text{ F})(6 \text{ H}) - 1)(4 \Omega)(5 \Omega)} \\
 &= \frac{-20 - j 54}{162 + j 7540} \text{ V} \\
 &= \frac{-20 - j 54}{162 + j 7540} \frac{162 - j 7540}{162 - j 7540} \text{ V} \\
 &= \frac{(-20 - j 54)(162 - j 7540)}{162^2 + 7540^2} \text{ V} \\
 &= \frac{-3240 + j 150800 - j 8748 - 407160}{26244 + 56851600} \text{ V} \\
 &= \frac{-410400 + j 142052}{56877844} \text{ V} \\
 &= \boxed{(-0.007215 + j 0.002497) \text{ V}}
 \end{aligned}$$

$$\begin{aligned}
 V_a &= V_b + V_{s2} \\
 &= (-0.007215 + j 0.002497) \text{ V} + j 2 \text{ V} \\
 &= \boxed{(-0.007215 + j 2.002497) \text{ V}}
 \end{aligned}$$

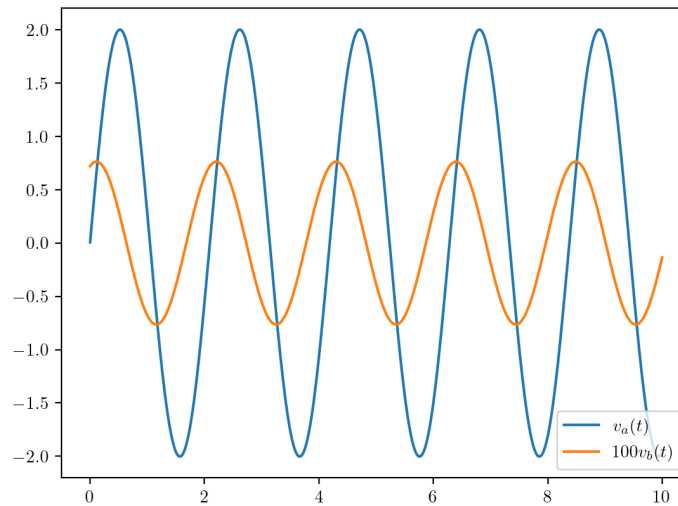
By calculating the amplitude A and phase θ , we can convert from complex number phasor notation $a + j b$ to either polar phasor notation $A \angle \theta$ or back into the time domain as $A \cos(\omega t + \theta)$. The amplitude and phase are calculated using:

$$A = \sqrt{a^2 + b^2}, \quad \theta = \arctan \frac{b}{a}$$

$$V_b = 0.007635 \angle -19.092^\circ$$

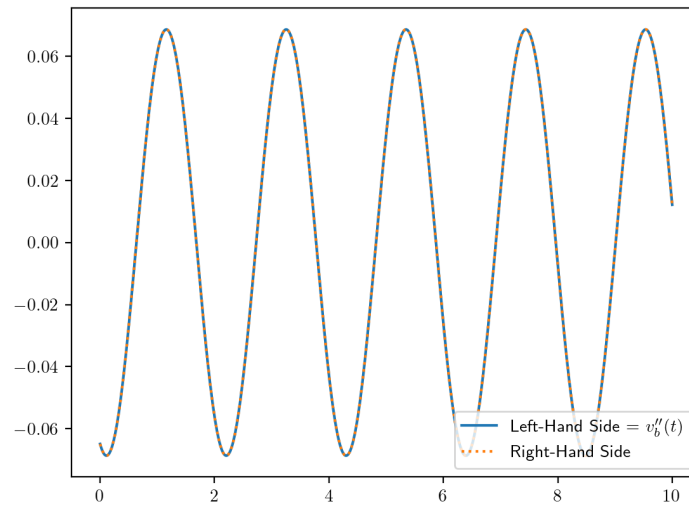
$$V_a = 2.002510 \angle -89.794^\circ$$

With the amplitude and phase calculated, it becomes very easy to plot the solution for $v_a(t)$ and $v_b(t)$:



3.4 Checking the phasor solution

At this point, we have derived the equations representing the circuit with and without phasors, and we have solved the circuit with phasors. To check our work, we can plug the solution obtained using phasors into the differential equation obtained without using phasors, and verify that the equation holds true (that the left-hand side equals the right-hand side). This can be done conveniently by plotting both the left- and right-hand sides:



As can be seen, the left- and right-hand sides are equal.

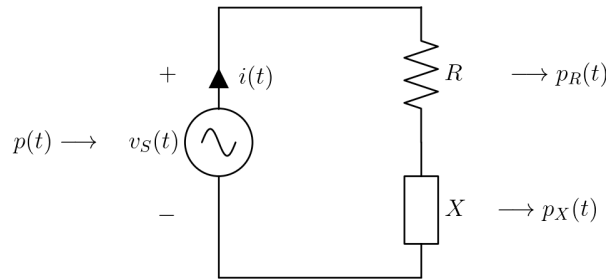
COMPLEX POWER AND THE POWER FACTOR

In the previous two chapters, we learned how complex numbers can simplify calculations involving voltage and current. We will see how complex numbers can also provide simplified power calculations. But first, in order to understand the nuances of AC power, we will see how power can be derived and quantified for a general AC circuit in the time domain.

4.1 Power derivations in the time domain

Consider an AC circuit with voltage source $v_S(t) = |V_S| \cos(\omega t + \theta_S)$ and impedance $Z = R + jX$. Such a circuit is useful because it can be used to model any series connection between one or more impedances and one or more voltage sources—as long as the voltage sources can be lumped together as one $v_S(t)$ and the impedances can be lumped together as one Z .

Figure 4.1



Using the approaches from the previous two chapters, we can determine the current through the circuit in the time domain:

$$i(t) = \frac{|V_S|}{\sqrt{R^2 + X^2}} \cos\left(\omega t + \theta_S - \arctan \frac{X}{R}\right)$$

Thus, the instantaneous power at time t is:

$$p(t) = v_S(t) i(t) = \frac{|V_S|^2}{\sqrt{R^2 + X^2}} \cos(\omega t + \theta_S) \cos\left(\omega t + \theta_S - \arctan \frac{X}{R}\right)$$

Applying the trigonometric identity $\cos \theta_a \cos \theta_b = (\cos(\theta_a - \theta_b) + \cos(\theta_a + \theta_b))/2$ gives us the following expression for $p(t)$. We see that $p(t)$ is a sinusoid, just like $v_S(t)$ and $i(t)$. However, $p(t)$ oscillates with twice the frequency of the voltage and current (2ω , instead of ω), and does not necessarily oscillate around zero. The time-invariant part of $p(t)$ is the “centerline” around which the sinusoid oscillates, which is also equal to the sinusoid’s time-averaged value denoted

by $\bar{p}$.

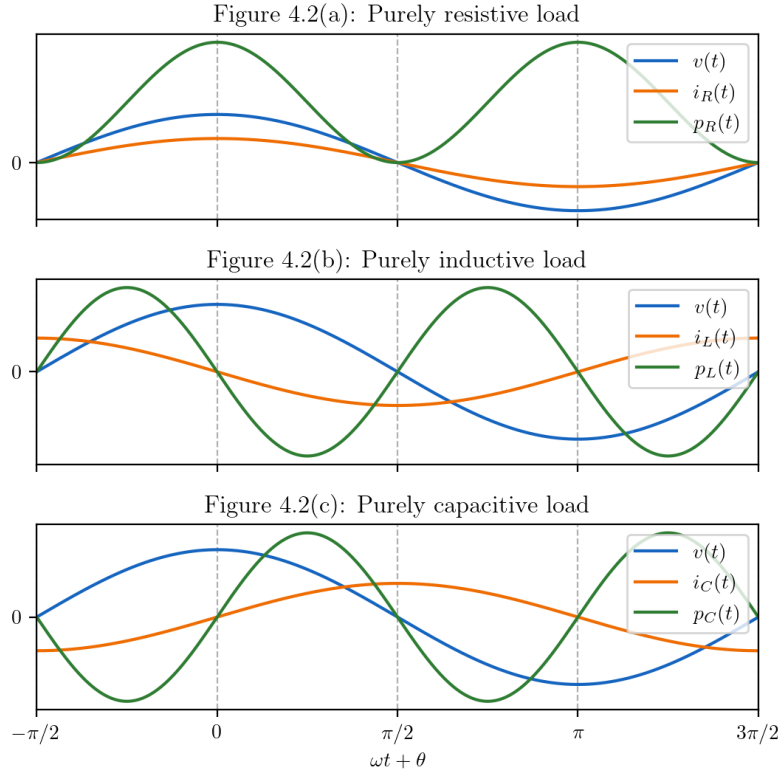
$$\begin{aligned}
 p(t) &= \frac{|V_S|^2}{\sqrt{R^2 + X^2}} \\
 &\times \frac{1}{2} \left(\cos \left(\omega t + \theta_S - \left(\omega t + \theta_S - \arctan \frac{X}{R} \right) \right) + \cos \left(2(\omega t + \theta_S) - \arctan \frac{X}{R} \right) \right) \\
 &= \frac{1}{2} \frac{|V_S|^2}{\sqrt{R^2 + X^2}} \cos \left(\arctan \frac{X}{R} \right) + \frac{1}{2} \frac{|V_S|^2}{\sqrt{R^2 + X^2}} \cos \left(2(\omega t + \theta_S) - \arctan \frac{X}{R} \right) \\
 &= \underbrace{\frac{1}{2} \frac{|V_S|^2}{R^2 + X^2} R}_{\bar{p}, \text{ time-invariant}} + \underbrace{\frac{1}{2} \frac{|V_S|^2}{\sqrt{R^2 + X^2}} \cos \left(2(\omega t + \theta_S) - \arctan \frac{X}{R} \right)}_{\text{time-variant}}
 \end{aligned} \tag{4.1}$$

We can derive $p(t)$ for a purely resistive AC circuit as follows, by setting $X = 0$. We can see that $p(t)$ is always greater than (or equal to) zero, as illustrated in Figure 4.2(a). Because resistor R always dissipates power and in the real world likely represents a device that can do useful work—such as a light bulb or laptop—we call this *real* or *active* power. In the $X = 0$ case, we can quantify our circuit's active power by its peak-to-peak value of $|V_S|^2/R$.

$$p(t) = \underbrace{\frac{1}{2} \frac{|V_S|^2}{R}}_{\bar{p}} + \underbrace{\frac{1}{2} \frac{|V_S|^2}{R} \cos(2(\omega t + \theta_S))}_{\text{time-variant part}}$$

Similarly, we can derive $p(t)$ for an AC circuit with only inductors and capacitors as follows, by setting $R = 0$. The average power $\bar{p}$ is now zero; $p(t)$ oscillates around zero as illustrated in figures 4.2(a) and 4.2(b). Recall from [Chapter 1](#) that inductive and capacitive components store energy. The rate at which the component stores or releases electrical energy is what we call its electric power—positive when converting *from* electrical energy and negative when converting back *to* electrical energy. When an AC voltage is applied to an inductive or capacitive component, it stores and releases energy in a cycle. The average power going into the component is equal to the average power going out of it over time. Power is exchanged back and forth between the AC voltage source and the inductor or capacitor, but none is dissipated or does any useful work. We call this *reactive* power. In the $R = 0$ case, we can quantify our circuit's reactive power by its peak-to-peak value of $|V_S|^2/X$.

$$p(t) = \underbrace{\frac{1}{2} \frac{|V_S|^2}{X} \cos \left(2(\omega t + \theta_S) + \frac{\pi}{2} \right)}_{\text{time-variant}}$$



But how do we quantify the power for a circuit with nonzero R and X ? There are two components—power being dissipated, and power being exchanged. Firstly, the power dissipated by the resistive load is:

$$p_R(t) = i^2(t)R = \frac{V_S^2}{R^2 + X^2} \cos^2\left(\omega t + \theta_S - \arctan \frac{X}{R}\right) R$$

Applying the trigonometric identity $\cos^2\theta = (\cos 2\theta + 1)/2$ gives us the following—the circuit's active power—whose peak-to-peak value is $V_S^2 R/(R^2 + X^2)$. Note that even though this is only active power and is only dissipated by the resistor, it is affected by X in addition to R .

$$p_R(t) = \underbrace{\frac{1}{2} \frac{V_S^2}{R^2 + X^2} R}_{\bar{p}_R, \text{ time-invariant}} + \underbrace{\frac{1}{2} \frac{V_S^2}{R^2 + X^2} R \cos\left(2\left(\omega t + \theta_S - \arctan \frac{X}{R}\right)\right)}_{\text{time-variant}} \quad (4.2)$$

And secondly, the power exchanged by the reactive load is:

$$\begin{aligned} p_X(t) &= p(t) - p_R(t) \\ &= \frac{1}{2} \frac{V_S^2}{R^2 + X^2} R + \frac{1}{2} \frac{V_S^2}{\sqrt{R^2 + X^2}} \cos\left(2(\omega t + \theta_S) - \arctan \frac{X}{R}\right) \\ &\quad - \frac{1}{2} \frac{V_S^2}{R^2 + X^2} R - \frac{1}{2} \frac{V_S^2}{R^2 + X^2} R \cos\left(2\left(\omega t + \theta_S - \arctan \frac{X}{R}\right)\right) \\ &= \frac{1}{2} \frac{V_S^2}{R^2 + X^2} \\ &\quad \times \left[\sqrt{R^2 + X^2} \cos\left(2(\omega t + \theta_S) - \arctan \frac{X}{R}\right) - R \cos\left(2\left(\omega t + \theta_S - \arctan \frac{X}{R}\right)\right) \right] \end{aligned}$$

To the expression enclosed in square brackets, we now apply the trigonometric identity $a \cos(x + \theta_a) + b \cos(x + \theta_b) =$

$c \cos(x + \theta_c)$, where $c = \sqrt{a^2 + b^2 + 2ab \cos(\theta_a - \theta_b)}$ and $\theta_c = \arctan((a \sin \theta_a + b \sin \theta_b)/(a \cos \theta_a + b \cos \theta_b))$.

$$x = 2(\omega t + \theta_S)$$

$$\begin{aligned} c &= \sqrt{\left(\sqrt{R^2 + X^2}\right)^2 + R^2 + 2\sqrt{R^2 + X^2}(-R) \cos\left(2(\omega t + \theta_S) - \arctan \frac{X}{R} - 2\left(\omega t + \theta_S - \arctan \frac{X}{R}\right)\right)} \\ &= \sqrt{2R^2 + X^2 - 2R\sqrt{R^2 + X^2} \cos\left(\arctan \frac{X}{R}\right)} \\ &= \sqrt{2R^2 + X^2 - 2R^2} \\ &= X \\ \theta_c &= \arctan \frac{\sqrt{R^2 + X^2} \sin\left(-\arctan \frac{X}{R}\right) - R \sin\left(-2 \arctan \frac{X}{R}\right)}{\sqrt{R^2 + X^2} \cos\left(-\arctan \frac{X}{R}\right) - R \cos\left(-2 \arctan \frac{X}{R}\right)} \\ &= \arctan \frac{-\sqrt{R^2 + X^2} \sin\left(\arctan \frac{X}{R}\right) + 2R \sin\left(\arctan \frac{X}{R}\right) \cos\left(\arctan \frac{X}{R}\right)}{\sqrt{R^2 + X^2} \cos\left(\arctan \frac{X}{R}\right) - R(2 \cos^2\left(\arctan \frac{X}{R}\right) - 1)} \\ &= \arctan \frac{-X + 2R \frac{XR}{R^2 + X^2}}{R - R\left(2 \frac{R^2}{R^2 + X^2} - 1\right)} \\ &= \arctan \frac{-X(R^2 + X^2) + 2R(XR)}{2R(R^2 + X^2) - 2RR^2} \\ &= \arctan \frac{R^2 - X^2}{2RX} \end{aligned}$$

Thus, reactive load's power is the following—the circuit's reactive power—whose peak-to-peak value is $V_S^2 X / (R^2 + X^2)$. Note that even though this is only reactive power and is only exchanged by the reactive load, it is affected by R in addition to X .

$$p_X(t) = \frac{1}{2} \frac{V_S^2}{R^2 + X^2} X \cos\left(2(\omega t + \theta_S) + \arctan \frac{R^2 - X^2}{2RX}\right)$$

Now we can determine the active and reactive power for any circuit like that in Figure 4.1.

4.2 Power as a phasor?

The previous section informed us of the nuances of AC power, but the calculations were tedious. We ought to be able to determine the active and reactive power of a circuit—or of any of its components—more easily. Perhaps if AC voltage and current calculations are simplified by working in the phasor domain, then so would AC power calculations. If $p(t) = v^2(t)/R$ for a purely resistive circuit, then surely the equivalent for a circuit that includes reactive components would be $P = V_S^2/Z$? Let's attempt to use this definition:

$$\begin{aligned} P &= \frac{V_S^2}{Z} \\ &= \frac{(|V_S| \angle \theta_S)^2}{\sqrt{R^2 + X^2} \angle \arctan \frac{X}{R}} \\ &= \frac{|V_S|^2 \angle 2\theta_S}{\sqrt{R^2 + X^2} \angle \arctan \frac{X}{R}} \\ &= \frac{|V_S|^2}{\sqrt{R^2 + X^2}} \angle \left(2\theta_S - \arctan \frac{X}{R}\right) \end{aligned} \tag{4.3}$$

If we were to naively convert this phasor into the time domain as follows, we would find that it does not match Equation (4.1) for $p(t)$ which we derived in the previous section. Firstly, because it oscillates around zero rather than around $|V_S|^2 R / 2(R^2 + X^2)$. Secondly, because its amplitude is $|V_S|^2 / \sqrt{R^2 + X^2}$ rather than $|V_S|^2 / 2\sqrt{R^2 + X^2}$. And thirdly, because it oscillates at natural frequency ω rather than 2ω . Clearly, this does not work—power cannot be represented by a phasor.

$$\begin{aligned} & \frac{|V_S|^2}{\sqrt{R^2 + X^2}} \cos\left(\omega t + 2\theta_S - \arctan \frac{X}{R}\right) \\ & \neq \underbrace{\frac{1}{2} \frac{|V_S|^2}{R^2 + X^2} R + \frac{1}{2} \frac{|V_S|^2}{\sqrt{R^2 + X^2}} \cos\left(2(\omega t + \theta_S) - \arctan \frac{X}{R}\right)}_{\text{From Equation 4.1}} \end{aligned}$$

However, this does not necessarily mean that AC power cannot be quantified by a complex number or that complex numbers cannot somehow be used to simplify AC power calculations. Note that active and reactive power do not depend on θ_S . In Equation (4.3), if we were to treat θ_S as if it were zero, we could write:

$$\begin{aligned} P &= \frac{|V_S|^2}{\sqrt{R^2 + X^2}} \angle \arctan \frac{X}{R} \\ &= \frac{|V_S|^2}{\sqrt{R^2 + X^2}} \cos\left(\arctan \frac{X}{R}\right) + j \frac{|V_S|^2}{\sqrt{R^2 + X^2}} \sin\left(\arctan \frac{X}{R}\right) \\ &= \frac{|V_S|^2}{\sqrt{R^2 + X^2}} \frac{R}{\sqrt{R^2 + X^2}} + j \frac{|V_S|^2}{\sqrt{R^2 + X^2}} \frac{X}{\sqrt{R^2 + X^2}} \\ &= \frac{|V_S|^2}{R^2 + X^2} R + j \frac{|V_S|^2}{R^2 + X^2} X \end{aligned}$$

Note how the real part $|V_S|^2 R / (R^2 + X^2)$ and the imaginary part $|V_S|^2 X / (R^2 + X^2)$ are the (peak-to-peak) values of the circuit's active and reactive power that we derived in the previous section. This means that we are en route to a way to use complex numbers to quantify AC power. In practice, it is not the entire expression $|V_S|^2 R / (R^2 + X^2) + j |V_S|^2 X / (R^2 + X^2)$ that we denote P , but just the real part $|V_S|^2 R / (R^2 + X^2)$. We denote the imaginary part $|V_S|^2 X / (R^2 + X^2)$ by Q , and the expression as a whole by complex number S . Thus, $S = P + jQ$. S is what we call *complex power*.

4.3 Complex power

By quantifying AC power as

$$S = \frac{|V_S|^2}{Z^*} \quad (4.4)$$

rather than as V_S^2 / Z , we can remove θ_S from our calculations and make them simpler. For example, for a circuit like that in Figure 4.1, we can calculate complex power S easily using only complex number division:

$$S = \frac{|V_S|^2}{R - jX} = \frac{|V_S|^2}{\underbrace{R^2 + X^2}_P} + j \frac{|V_S|^2}{\underbrace{R^2 + X^2}_Q} X$$

From Equation (4.4), we can derive the following—the AC power equivalent of the formula $P = I^2 R$.

$$S = \frac{|V_S|^2}{Z^*} = \frac{|V_S|^2}{Z^*} \frac{Z}{Z} = \frac{|V_S|^2 Z}{|Z|^2} = \left(\frac{|V_S|}{|Z|} \right)^2 Z \Rightarrow \boxed{S = |I|^2 Z} \quad (4.5)$$

From which we can derive the AC power equivalent of $P = V_S I$:

$$S = |I|^2 Z = |I|^2 \frac{V_S}{I} = |I|^2 \frac{V_S}{I} \frac{I^*}{I^*} = |I|^2 \frac{V_S I^*}{|I|^2} \Rightarrow \boxed{S = V_S I^*} \quad (4.6)$$

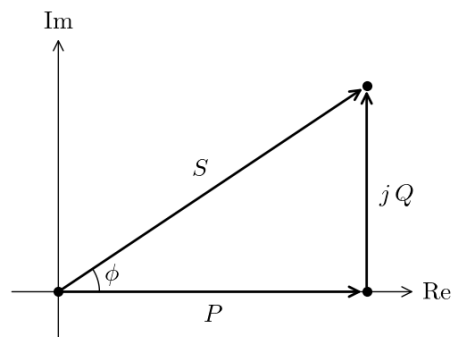
Looking at complex power in polar form, its magnitude $|S|$ is what we call *apparent power*, and its angle $\phi = \arctan Q/P$ is the same as the phase angle $\arctan X/R$ by which the current lags (or leads) the voltage. The following table summarizes the values taken on by P , Q , S , and ϕ for different types of load.

Load	Active power P	Reactive power Q	Complex power S	Phase angle ϕ
Purely resistive	> 0	$= 0$	$= P$	$= 0$
Purely inductive	$= 0$	> 0	$= jQ$	> 0
Purely capacitive	$= 0$	< 0	$= jQ$	< 0

4.4 The power triangle

In *What is a Phasor?*, we saw how a triangle could be used to depict a phasor (and thus any complex number) in the complex plane. Accordingly, complex power is often depicted as a *power triangle*, which aids in remembering the relationships between P , Q , S , and ϕ .

Figure 4.3



Even though power $p(t)$ is a sinusoid, and a phasor rotating in the complex plane traces out a sinusoid, S is not a phasor and would not form $p(t)$ if rotated about the origin. However, if $S - P/2$ were rotated about $P/2$, it would correctly trace out $p(t)$ as illustrated in Figure 4.4(a). Figures 4.4(b), 4.4(c), and 4.4(d) are for purely resistive, inductive, and capacitive loads and correspond to figures 4.2(a), 4.2(b), and 4.2(c).

Figure 4.4(a): Resistive and inductive load

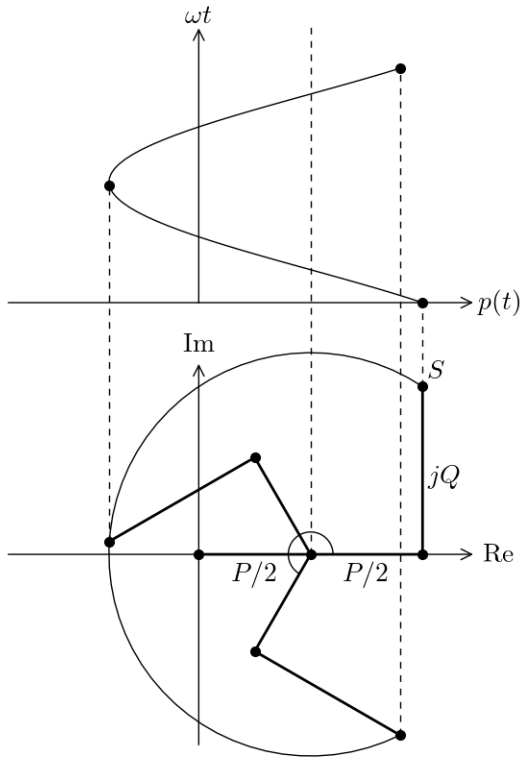


Figure 4.4(b): Purely resistive load

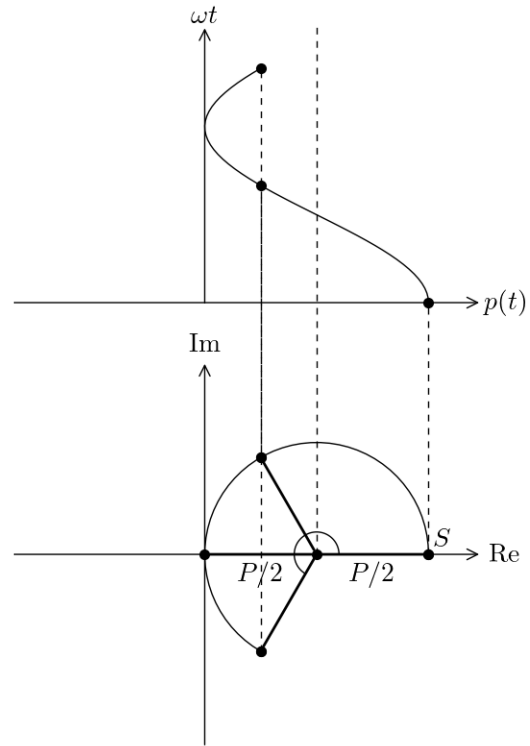


Figure 4.4(c): Purely inductive load

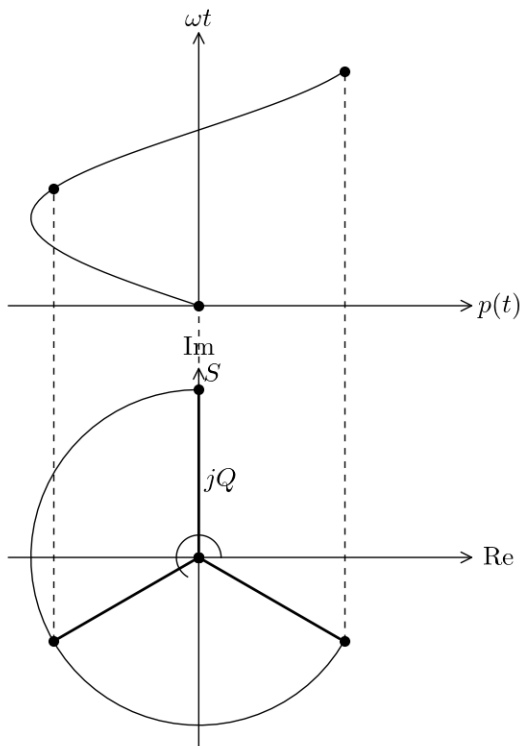
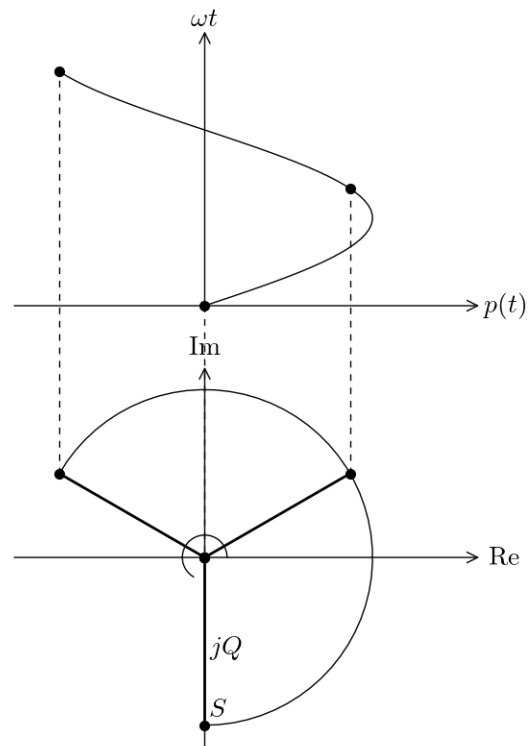


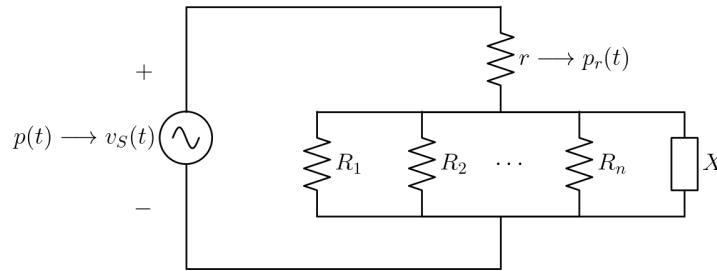
Figure 4.4(d): Purely capacitive load



4.5 The power factor

This section introduces the concept of the power factor. The power factor is a general concept in AC systems, but is particularly consequential for the electrical grid and its users. So, understanding why the concept of the power factor exists and its implications first requires a high-level understanding of the electrical grid. The electrical grid can be roughly modeled at a very high level by the circuit shown in Figure 4.5. Voltage source v_S represents a generating station. The resistances of value R_i and the reactance X represent loads connected to the grid in parallel, from multiple grid users. Whereas the resistances of value R_i may represent loads such as light bulbs or laptops, X represents a reactive load, such as an inductive ballast in a fluorescent light, or a large industrial induction motor. Resistance r represents the resistance of transmission lines between the generating station and the grid loads. Note that liberties have been taken at every part of this model. In reality, grids are complex circuits with multiple generating stations and loads, all connected with a network of transmission and distribution lines. Generating stations cannot just be represented by voltage sources, and more accurate transmission line models incorporate reactances. Nonetheless, the circuit shown in Figure 4.5 is a sufficient model for the purposes of understanding the power factor.

Figure 4.5



Using the formulae for impedances in series and in parallel, the total impedance looking into the grid loads from the generating station is as follows. We can see that the presence of reactive load X reduces the magnitude of the impedance $|Z|$.

$$Z = r + \frac{1}{n \cdot \frac{1}{R} + \frac{1}{jX}}$$

Adapting Equation (4.1), the power $p(t)$ from the generating station is as follows. The average and peak-to-peak values of $p(t)$ are $\bar{p}$ and $|V_S|^2/|Z|$, respectively.

$$p(t) = \underbrace{\frac{1}{2} \frac{|V_S|^2}{|Z|^2} \text{Re}(Z)}_{\bar{p}} + \frac{1}{2} \frac{|V_S|^2}{|Z|} \cos\left(2(\omega t + \theta_S) - \arctan \frac{\text{Im}(Z)}{\text{Re}(Z)}\right)$$

And adapting Equation (4.2), the power $p_r(t)$ dissipated by resistance r is as follows. The average and peak-to-peak values of $p_r(t)$ are $\bar{p}_r$ and $|V_S|^2 r / |Z|^2$, respectively.

$$p_r(t) = \underbrace{\frac{1}{2} \frac{|V_S|^2}{|Z|^2} r}_{\bar{p}_r} + \frac{1}{2} \frac{|V_S|^2}{|Z|^2} r \cos\left(2\left(\omega t + \theta_S - \arctan \frac{\text{Im}(Z)}{\text{Re}(Z)}\right)\right)$$

If $|Z|$ is reduced due to the presence of reactive load X , then this will increase the peak-to-peak value of $p(t)$, the peak-to-peak value of $p_r(t)$, and the average value of $p_r(t)$. This has a number of consequences. Firstly, the generating station V_S must be able to produce higher peak instantaneous power. Secondly, the transmission line r must be able to dissipate a higher average power without overheating. And thirdly, because the transmission line r is dissipating more power, its voltage drop is higher. Thus, the generating station must operate at higher voltage and power in order to maintain the same voltage and power for each connected load. Thus, reactive loads tend to incur higher infrastructure costs, in the form of

higher-capacity generators and transmission lines, and higher operating costs. This extra demand on the grid would exist even if reactive load X were instead a resistive load—but, with a reactive load, the extra demand is unnecessary. With a reactive load, extra power is being transferred back and forth needlessly between the load and the generating station. Whenever the reactive load stores energy, the generating station must be providing it, and whenever the reactive load releases energy, the generating station must be absorbing it.

It would not be fair for the generating station or grid operator to pay for the higher infrastructure costs that reactive loads require, nor would it be fair for the generating station to pay for the higher operating costs. So, the cost of the extra demand on the grid is often passed onto the grid user who is responsible for it according to their power meter—for example, an industrial utility customer. This is done by having grid users' power meters report the apparent power $|S|$, rather than just the active power P , over time. Grid users can reduce their electricity costs, without reducing their active power usage, by reducing their reactive loads. This reduces their reactive power Q , which in turn reduces $|S|$.

Note

If your power factor is low, you are paying extra to have power needlessly transmitted back and forth between your reactive loads and the grid's generating stations, potentially over hundreds of kilometers of transmission lines, sixty times a second.

The ratio between P and $|S|$ is what is known as the power factor (PF):

$$\text{PF} = \frac{P}{|S|}$$

The power factor ranges from 0 for a purely reactive load to 1 for a purely resistive load. However, a negative power factor can arise when the “load” (the grid user) is actually supplying active power rather than consuming it. The power factor is described as *lagging* for a more inductive load with $\phi > 0$, or *leading* for a more capacitive load with $\phi < 0$. Because the power factor is the ratio between the adjacent side and hypotenuse of the power triangle, it can be reformulated as $\cos \phi$.

Increasing the power factor reduces electricity costs. This can be done by choosing resistive loads over reactive loads where possible (e.g., LEDs over fluorescent lights), by implementing *power factor correction (PFC)*, or by choosing loads that have power factor correction built-in. Implementing power factor correction is important when installing loads that necessarily have a low power factor (such as an induction motor) or when a grid user seeks to reduce the costs incurred by low-power-factor loads that they already have and want to keep using (such as fluorescent lights).

4.6 Power factor correction

Most loads with a low power factor are inductive. Because of this, their lagging power factor (inductive reactance) can be counteracted using loads with a leading power factor—such as synchronous motors, or large banks of capacitors whose sole purpose is to do power factor correction. The reactive power exchanged by a capacitor is exactly 180° out-of-phase with an inductive load, meaning that when the inductive load stores energy, the capacitor discharges to provide that energy, and when the inductive load releases energy, the capacitor charges to absorb that energy. The required capacitance C can be calculated such that the power exchanged by the capacitor is exactly opposite to that exchanged by the inductive load. This can be done by calculating what value of C sets the total reactance to zero. Let the impedance of the inductive load be $\text{Re}(Z_L) + j \text{Im}(Z_L)$ rather than just ωL to account for the fact that its equivalent circuit likely includes resistances. The capacitor can be placed in series or in parallel with the inductive load, but if in parallel, the power-factor-corrective capacitors do not need to be co-located with a facility's inductive loads.

For a capacitor C in series with inductive load Z_L :

$$\begin{aligned}Z_{\text{tot.}} &= Z_L + Z_C = \text{Re}(Z_L) + j\text{Im}(Z_L) + jX_C \\ \text{Im}(Z_{\text{tot.}}) &= 0 \implies \text{Im}(Z_L) + X_C = 0 \\ &\implies \text{Im}(Z_L) - \frac{1}{\omega C} = 0 \\ &\implies C = \frac{1}{\omega \text{Im}(Z_L)}\end{aligned}$$

For a capacitor C in parallel with inductive load Z_L :

$$\begin{aligned}Z_{\text{tot.}} &= \frac{1}{Z_L} + \frac{1}{Z_C} = \frac{1}{\text{Re}(Z_L) + j\text{Im}(Z_L)} + \frac{1}{jX_C} = \frac{\text{Re}(Z_L) - j\text{Im}(Z_L)}{|Z_L|^2} - j\frac{1}{X_C} \\ \text{Im}(Z_{\text{tot.}}) &= 0 \implies -\frac{\text{Im}(Z_L)}{|Z_L|^2} - \frac{1}{X_C} = 0 \\ &\implies -\frac{\text{Im}(Z_L)}{|Z_L|^2} - \frac{1}{-1/\omega C} = 0 \\ &\implies C = \frac{\text{Im}(Z_L)}{\omega |Z_L|^2}\end{aligned}$$

The capacitor is taking the place of the generating station in terms of providing and absorbing the extra energy that the inductive load must store and release. But, because this extra energy no longer needs to be transmitted over long distances to and from a generating station, the infrastructure and other costs are reduced and no longer passed on to the grid user. The grid users with the highest cost savings from power factor correction are industrial plants with many, large induction motors. Residential utility customers are usually not metered or charged for their reactive power, and so would not typically benefit from power factor correction. There do exist small capacitive devices that plug directly into wall outlets and supposedly do power factor correction for residents, but they would save nothing for most residential utility customers—that is, if they do any power factor correction at all.